Maximum Blackouts That Keep Every Lattice Rectangle Identifiable

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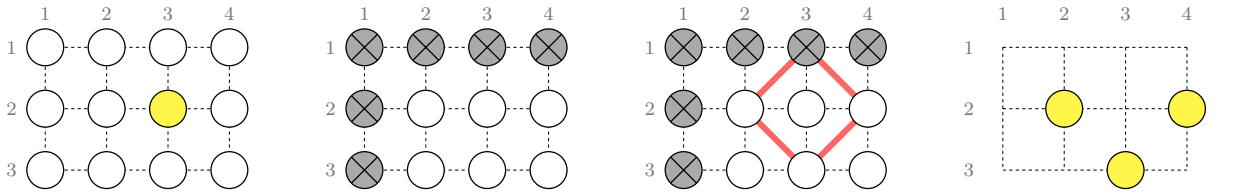
Abstract

How many points of a grid can we black out while every rectangle, tilted or not, remains determined by its white corners? This is Kagey's Problem 001. A row-column graph and Mantel's theorem bound the maximum by the width plus the integer part of a quarter of the squared height. We attain this bound at every height for all sufficiently large widths by spacing columns with two black points between runs black in alternating boundary rows. We determine the maximum on up to six rows at every width, and on seven rows from width 24 on. The five-row result refutes our earlier guess that the maximum is the sum of the sides. We count the three-row optima exactly and find the growth rate on four rows. At fixed height, the maximum minus the width is eventually periodic, and the optimum counts have a rational generating function. The maxima and counts are tabulated in A397315 and A399580 of the On-Line Encyclopedia of Integer Sequences. Some values require computer checks.

1 Introduction

Start with an $h \times m$ grid and black out some of its points. The game master chooses a rectangle (tilted ones are allowed) and tells you which of its corners are white. Can you recover all four corners? Kagey [21, 19] asks how many points can be blacked out while keeping the answer unique for every rectangle. Such a set is a *valid blackout* (Definition 1). One round is shown in Figure 1.

Write $f(h, m)$ for the maximum blackout size, and $a_h(m)$ for the number of optima. The OEIS entry A397315 [34] conjectured $f(3, m) = m+2$ and $f(h, m) = h+m$ for $\min(h, m) \geq 4$, with exceptions $f(4, 4) = 7$ and $f(4, 5) = 8$. The first was proved in Lean by the Omega Institute, whose pull request was merged on 15 September 2026 [35]. Our independent proof combines the row bound (Theorem 8) and the boundary cross (Lemma 17). The second



(a) the point $(3, 2)$ (b) our blackout (c) the rectangle (d) what we are shown

Figure 1: One round on a largest valid blackout of the 3×4 grid (Section 5). Points are (column, row). Gray crossed circles are black. Panel (c) shows the red rectangle, and (d) its yellow corners. The three yellow corners fix the fourth by Lemma 4.

guess breaks at 5×8 (Theorem 40). Throughout, put $q_h = \lfloor h^2/4 \rfloor$. Computer-assisted claims are marked, and the supplement to this paper lists the evidence for each.

The graph bound is sharp at every height once the grid is wide enough. To attain it, we separate the columns containing two black points and alternate the intervening black points between the boundary rows. The separation lets us check tilted rectangles near one double column at a time. The main results follow, with proof locations in parentheses.

Theorem A (narrow grids)**.**

(a) We have $f(h, m) \geq h + m - 1$*, and* $f(h, m) \leq m + q_h$ *if* $m \geq 3$*.* (Lemma 17 and Theorem 8, Section 4.)

(b) We have $f(2, m) = m + 1$ *and* $f(3, m) = m + 2$ *for* $m \geq 3$*,* $f(4, m) = m + 4$ *for* $m \geq 6$*, and* $f(5, 5) = 10$*.* (Theorem 5, Section 5, Theorem 31, and Theorem 46(a).)

(c) (Computer-assisted, with checked certificates for $f(5, 6)$ *and* $f(5, 7)$*, and the last claim checked in Lean.) We have* $f(4, 4) = 7$*,* $f(4, 5) = 8$*,* $f(5, 6) = 11$*,* $f(5, 7) = 12$*, and* $f(5, m) = m + 6$ *for* $m \geq 8$*.* (Corollary 32 and Theorem 40(a), (b).)

(d) (Computer-assisted for $m \leq 14$*, with checked certificates for the upper bounds, and checked in Lean for* $m \geq 15$*.) We have* $f(6, m) = m + 7$ *for* $8 \leq m \leq 11$*,* $f(6, m) = m + 8$ *for* $12 \leq m \leq 14$*, and* $f(6, m) = m + 9$ *for* $m \geq 15$*.* (Theorem 43.)

(e) (Computer-assisted, with checked certificates.) We have $f(6, 6) = 12$*,* $f(6, 7) = 13$*,* $f(7, 7) = 14$*, and* $f(6, 8) = 15$*.* (Theorem 45.)

(f) (Computer-assisted.) We have $f(7, m) = m + 12$ *for every* $m \geq 24$*.* (Theorem 44.)

With (e) and transposition, $f(h, m)$ is known for every $h \leq 6$ and every m. For (c) and (d), the collar theorem reduces each infinite family to one validity check (Theorem 39). Lemma 54 justifies the SAT bounds.

Theorem B (three-row optima)**.**

(a) For $m \geq 4$ the counts $a_3(m)$ (A401223) satisfy $a_3(m) = 4(m-1)(3m+2)\,3^{m-5} + E_m$, where $E_m > 0$ counts the optima with a column black in rows 1 and 3 (A398448), and $E_m = O(m^2(1+\sqrt{3})^m)$. So $a_3(m) \sim \frac{4}{81} m^2 3^m$, and a fraction tending to 4/27 of the $m^2 3^{m-1}$ axis-valid candidates is valid. The series $\sum_{m\geq 3} a_3(m) z^m$ is rational, and its denominator divides $(1-3z)^3(1-2z-2z^2)^3(1+2z^2)^3$. (Theorem 24, Proposition 19(c), the last claims of Theorem 26, and Theorem 25(a), Section 5.)

(b) (Computer-assisted, exact computer algebra.) This is the reduced denominator, and $E_m \sim \frac{19+11\sqrt{3}}{432} m^2 (1+\sqrt{3})^m$. (Theorem 25(b), proved in Section 5.2, and Theorem 26.)

Theorem C (four rows)**.** *If $m \geq 4$, no valid blackout of the $4 \times m$ grid hides all corners of a rectangle* (Theorem 33)*. (Computer-assisted.) The series $\sum_{m\geq 6} a_4(m) z^m$ of the counts in A401224 is rational with reduced denominator of degree 420, and $a_4(m) \sim c\, m^4 \rho_4^m$ with $c \approx 8.1611 \times 10^{-5}$ and $\rho_4 = 3.35199\ldots$, an algebraic number of degree 20* (Proposition 36, Theorem 37, and Corollary 38, Section 6).

Theorem D (every height at large widths)**.** *For every $h \geq 3$ we have $f(h,m) = m + q_h$ once $m \geq q_h + (q_h - 1)(2W_h + 1)$, where $W_h = 1 + \lfloor (h-1)^2/4 \rfloor$.* (Theorem 47, Section 8.)

Theorem E (every fixed height)**.** *Fix $h \geq 1$ and write a blackout as the word $B_1 \cdots B_m$ of the black rows of its columns. The valid words form a regular language. If $h \geq 2$, then $f(h,m) - m$ is eventually periodic in m, and $\sum_{m\geq 0} a_h(m) z^m$ is rational.* (Theorems 49 and 50, Section 9.)

Corollary 15 proves the conjecture in A338885 that its row n has largest entry A033638(n).

Prior work. Choosing white points to distinguish rectangles is a test-cover problem [4, 13, 22, 29] (Corollary 3). If we only forbid fully black axis-aligned rectangles, we get the Zarankiewicz problem [14, 23, 32]. To distinguish all axis-aligned rectangles, the row-column graph must avoid both four- and six-cycles (Lemma 7). Such graphs, including those with unequal parts, are studied by de Caen and Székely [8, 9]. With n vertices on each side, the largest ones have $\Theta(n^{4/3})$ edges. Incidence graphs of generalized quadrangles attain this order [5], and Hoory [17] and Neuwirth [30] bound it from above. Lazebnik, Ustimenko, and Woldar [24] construct dense graphs of every even girth. Neuwirth's edge bounds [30, Theorem 1(ii) and Proposition 17] give our row bound in graph form. Our proof using Mantel's theorem [25] also classifies and counts equality cases. We then check the tilted rectangles. At fixed height, we count column words using transfer matrices [1, 10, 26], [11, Proposition V.6]. Theorem 1(3) of Guilhot, Little, and Parkinson [15] also yields Theorem E once regularity is proved.

Outline. Sections 2 to 4 give the definitions and graph bounds. Sections 5 to 7 treat narrow grids, followed by large widths in Section 8, fixed height in Section 9, and squares in Section 10. Sections 11 and 12 give the verification and open questions.

2 Conventions

We record the black points column by column, using the following notation.

(1) An $h \times m$ grid has h rows, m columns, and point set V. A $2 \times m$ grid is a *strip*. The point (i, j) lies in column i and row j, with $(1, 1)$ at the top left (Figure 1(a)).

(2) A *rectangle* has four grid points as corners and positive area. We write $C(R)$ for its corner set. It is *axis-aligned* if its sides follow the grid lines, and *tilted* otherwise.

(3) A *blackout* is the set $S \subseteq V$ of black points. Its *kept set* is $K = V \setminus S$, and the *presentation* $P_S(R) = C(R) \setminus S$ is what the rectangle shows, namely its white corners.

(4) Column i has *black rows* $B_i = \{j : (i, j) \in S\}$, and is *empty*, a *singleton*, a *double*, or a *triple* when $|B_i|$ is 0, 1, 2, or 3, and *full* when $|B_i| = h$. The *pair set* F is the set of row pairs contained in some B_i, and $q_h = \lfloor h^2/4 \rfloor$ is A002620(h).

Definition 1 (valid blackout)**.** A blackout S is *valid* if distinct rectangles have distinct presentations. We write $f(h, m)$ for the largest size of a valid blackout of the $h \times m$ grid, and $a_h(m)$ for the number of valid blackouts of that size. We count them with their coordinates (reflections and rotations are not identified).

Transposing the grid gives $f(h, m) = f(m, h)$ and $a_h(m) = a_m(h)$. There are no rectangles at width zero or height one, so we set $f(h, 0) = 0$, $a_h(0) = 1$, $f(1, m) = m$ and $a_1(m) = 1$. A rectangle is allowed to show no corners, but at most one can do so in a valid blackout.

Lemma 2 (difference sets)**.** *For sets A, B, and S, we have $A \setminus S = B \setminus S$ if and only if $A \triangle B \subseteq S$.*

Proof. The two differences agree exactly when every point in just one of A and B belongs to S. □

Corollary 3 (hitting sets)**.** *A blackout S is valid if and only if its kept set K meets $C(R_1) \triangle C(R_2)$ for every pair of distinct rectangles R_1, R_2.*

Apply Lemma 2 to the corner sets. Keeping an extra point cannot create an ambiguity, so every subset of a valid blackout is valid. Thus ruling out size k rules out every larger size.

Lemma 4 (three corners suffice)**.** *Three corners of a rectangle determine it, so distinct rectangles with the same presentation show at most two corners.*

Proof. Three corners form a right triangle, whose only right angle is at the corner q adjacent to the other two, p and r. So the fourth corner is $p + r - q$. □

3 Strips

On two rows every rectangle is axis-aligned. To see this, take a lowest corner of a tilted rectangle. Its two neighbors lie strictly above it, and the opposite corner lies above both, so the rectangle meets at least three rows. We can therefore identify strip rectangles by their two columns.

Theorem 5 (strip theorem)**.** *Let $m \geq 3$.*

(a) A blackout of the $2 \times m$ strip is valid if and only if at most one of its columns is full.

(b) We have $f(2, m) = m + 1$, and the maximum blackouts are those with one full column and one black point in every other column.

(c) There are $a_2(m) = m \cdot 2^{m-1}$ maximum blackouts, so row 2 of A399580 is A001787 from $m = 3$ on.

For $m = 2$ there is one rectangle, so $f(2, 2) = 4$ and $a_2(2) = 1$.

Proof. (a) Two distinct rectangles share at most one column. Their difference set contains both points of at least two columns. Conversely, every pair of columns p, q is a difference set. Choose a third column z (this is where $m \geq 3$ is used) and take the rectangles on $\{z, p\}$ and $\{z, q\}$. By Corollary 3, validity is thus equivalent to keeping a point in every pair of columns, or having at most one full column.

(b, c) We must keep at least one point in all but one column, so $|S| \leq 2m-(m-1) = m+1$. Equality requires one full column and exactly one black point in each other column. There are m choices for the full column, then two choices in each of the remaining $m - 1$ columns. □

Proposition 6 (orbits and kings)**.** *For $m \geq 3$ the number o_m of maximum blackouts of the $2 \times m$ strip up to the row reflection and the column reversal is $m\,2^{m-3} + [m \text{ odd}]\,2^{(m-3)/2}$. It equals A322284$(m - 1)$, which counts $m - 1$ nonattacking kings on a $2 \times (2m - 2)$ board up to symmetry.*

Proof. Apply Burnside's lemma to the four symmetries. The identity fixes all $m2^{m-1}$ optima, and swapping the rows fixes none. A column reversal or half turn can fix an optimum only when m is odd and the full column is central. Each then has $2^{(m-1)/2}$ fixed optima, since the singleton columns come in pairs. Dividing their sum by four gives the formula.

Put $n = m - 1$. Nonattacking kings on the $2 \times 2n$ board are columns $c_1 < \cdots < c_n$ with $c_{j+1} \geq c_j + 2$, and any rows r_j. The gaps $c_1 - 1$, $c_{j+1} - c_j - 2$, and $2n - c_n$ are nonnegative and sum to 1, so exactly one, say the eth, is 1, and $c_j = 2j - 1 + [j \geq e]$. Map this configuration to the optimum with full column e and singleton rows $r_1, \ldots, r_n$ from left to right. This is a bijection. It commutes with row reflection and with reversal, which sends e to $n + 2 - e$ and c to $2n + 1 - c$ and reverses the order of the row labels, since $2n + 1 - c_{n+1-j} = 2j - 1 + [j \geq n + 2 - e]$. So the orbits correspond. □

4 The row-column graph and general bounds

A blackout S is *axis-valid* when it distinguishes axis-aligned rectangles. Call a rectangle *weak* if at most two corners show. In the *row-column graph* G_S, join row r_j to column c_i for each black point (i, j).

Lemma 7 (axis-aligned rectangles and short cycles)**.** *If G_S has no four-cycle and no six-cycle, then S is axis-valid. The converse holds for $m \geq 3$.*

Proof. For $m \geq 3$, a four-cycle $r_j\, c_i\, r_{j'}\, c_{i'}$ gives indistinguishable rectangles $\{i, w\} \times \{j, j'\}$ and $\{i', w\} \times \{j, j'\}$ using a third column w. A six-cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ gives $\{x, y\} \times \{a, b\}$ and $\{y, z\} \times \{a, c\}$, both black except possibly at (y, a). Conversely, axis-aligned $R_1 \neq R_2$ with equal presentations have $C(R_1) \triangle C(R_2) \subseteq S$. They cannot share opposite corners, so by symmetry any common corners lie in one row a. With none, R_1 is fully black. With two, (x, a) and (y, a), the other rows $b \neq c$ give a black four-cycle $r_b\, c_x\, r_c\, c_y$. With just (x, a), write $R_1 = \{x, y\} \times \{a, b\}$ and $R_2 = \{x, z\} \times \{a, c\}$ to obtain $r_a\, c_y\, r_b\, c_x\, r_c\, c_z$. □

Theorem 8 (row bound)**.** *Let $m \geq 3$. Every axis-valid blackout of the $h \times m$ grid has at most $m + q_h$ points, and at most $h + q_m$ if also $h \geq 3$. Hence $f(h, m) \leq m + q_h$.*

Join each nonempty column's black rows B_i by a tree T_i. We apply Mantel's theorem to their union H.

Lemma 9 (row graph)**.** *Let $m \geq 3$ and let S be axis-valid. For any choice of the trees, H is simple and triangle-free, and $e(H) = |S| - t$, where t is the number of nonempty columns.*

Proof. A repeated edge of H would put the same row pair in two columns, giving a four-cycle in G_S. Lemma 7 rules this out, so $e(H) = \sum(|B_i| - 1) = |S| - t$. Suppose H contains a triangle $r_a r_b r_c$. Its edges cannot all belong to one tree. If two columns i, i' supply them, two of a, b, c belong to $B_i \cap B_{i'}$, again giving a four-cycle. If three columns supply them, they give the six-cycle $r_a\, c_i\, r_b\, c_{i'}\, r_c\, c_{i''}$. Both possibilities contradict Lemma 7. □

Lemma 10 (Mantel [25])**.** *Let H be a simple triangle-free graph on h vertices, let a vertex of maximum degree Δ have neighbor set Γ, and let Ω be the other $h - \Delta$ vertices, spanning $e(\Omega)$ edges. Then*

$$e(H) = \sum_{b \in \Omega} d(b) - e(\Omega) \leq (h - \Delta)\Delta - e(\Omega) \leq q_h.$$

Proof. The set Γ is independent, so every edge meets Ω. The degree sum over Ω counts its internal edges twice. Subtract them once, then use $d(b) \leq \Delta$ and $(h - \Delta)\Delta \leq q_h$. □

Proof of Theorem 8. Combining Lemmas 9 and 10 gives $|S| = t + e(H) \leq m + q_h$. The need for a third column enters only through Lemma 7. Transposing therefore gives $|S| \leq h + q_m$ when $h \geq 3$. □

Theorem 11 (equality structure)**.** *Let $h \geq 4$ and $m \geq 3$. A blackout of the $h \times m$ grid with $m + q_h$ points is axis-valid if and only if*

(a) every column is a singleton or a double,

(b) the q_h doubles have distinct row pairs, and

(c) these pairs form a balanced complete bipartite graph on the rows.

There are $p_h\,(m)_{q_h}\,h^{m-q_h}$ such blackouts, where $(m)_q$ is the falling factorial and the number p_h of balanced bipartitions of the rows is $\binom{h}{\lfloor h/2 \rfloor}$, halved for even h.

Proof. Equality in the row-bound proof forces $t = m$, $e(\Omega) = 0$, and $(h - \Delta)\Delta = q_h$, with all of Ω joined to all of Γ. Thus H is the graph in (c). If $|B_i| \geq 3$, its tree T_i has a vertex v with neighbors u, w. Since $h \geq 4$, the part containing v has another vertex z. Replacing vw by uw preserves uz, wz and creates triangle uwz, contradicting Lemma 9. This proves (a), and the tree edges, now the row pairs of doubles, give (b) and (c). Conversely, (a) and (b) exclude four-cycles. A six-cycle $r_a\,c_x\,r_b\,c_y\,r_c\,c_z$ needs doubles $\{a, b\}$, $\{b, c\}$, and $\{a, c\}$, contradicting (c). Apply Lemma 7. To count, choose the bipartition, assign edges to distinct columns, and choose the remaining singleton rows. □

Corollary 12 (small widths)**.** *Let $h \geq 4$ and $3 \leq m < q_h$. Then $f(h, m) \leq m + q_h - 1$. In particular $f(5, 5) \leq 10$.*

Proof. Attaining $m + q_h$ would require $q_h \leq m$ double columns by Theorem 11, contrary to the hypothesis. The bound in Theorem 8 therefore improves by one. For five rows, $q_5 = 6$. □

Put $W_h = 1 + \lfloor (h-1)^2/4 \rfloor$, so $W_3 = 2$, $W_4 = 3$, and $W_5 = 5$.

Lemma 13 (leftmost corner)**.** *A tilted rectangle R has a unique corner $\ell = (x, r)$ in its smallest column, and a unique corner in its smallest row. The sides from ℓ are $(a, -b)$ and (c, d) with positive integers a, b, c, d and $ac = bd$, and the corners are*

$$(x, r), \qquad (x + a,\, r - b), \qquad (x + c,\, r + d), \qquad (x + a + c,\, r - b + d). \tag{1}$$

Its rows are $r - b$ to $r + d$, and its span *is $a + c$.*

Proof. No side is vertical. Two corners in the leftmost column x would be opposite. Their midpoint would force the other two into that column as well. The same argument applies to rows. The two sides from ℓ therefore point right, one up and one down. Perpendicularity gives $ac = bd$. □

Lemma 14 (span bound)**.** *Every tilted rectangle of the $h \times m$ grid has span $a + c \leq W_h$.*

Proof. The rectangle fits in h rows, so $b + d \leq h - 1$. Also $(a - 1)(c - 1) \geq 0$, whence $a + c \leq bd + 1 \leq W_h$, using the integer bound $bd \leq (b + d)^2/4$. □

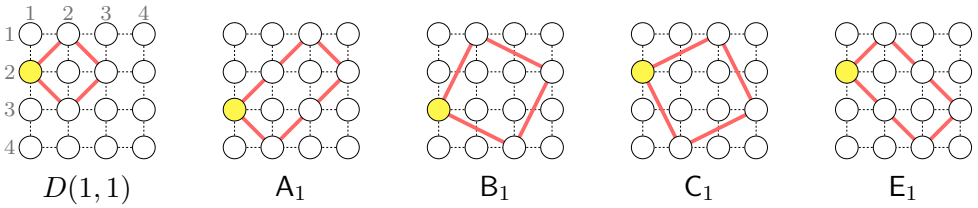


Figure 2: The tilted rectangles of the $4 \times m$ grid, up to translation. Each is drawn in a 4×4 window, and its leftmost corner ℓ is yellow.

Thus W_h is A033638$(h-1)$. For $h \geq 3$ and $m > W_h$, the sides $(1, -b)$ and (bd, d) attain it when $b + d = h - 1$ and $|b - d| \leq 1$. Here a rectangle's *height* and *width* mean its row and column ranges (the distances between its extreme coordinates).

Corollary 15 (the conjecture in A338885)**.** *For $n \geq 2$, the largest width of a tilted lattice rectangle of height n is $\lfloor n^2/4 \rfloor + 1$. So row n of A338885 has largest entry A033638(n).*

Proof. In (1), set the height $b + d = n$. As in the span bound, the width is $a + c \leq bd + 1 \leq \lfloor n^2/4 \rfloor + 1$. Choose sides $(1, -b)$ and (bd, d) with $b = \lfloor n/2 \rfloor$ and $d = \lceil n/2 \rceil$ to attain it. □

The *diamond* $D(i, j)$ is the square with corners $(i, j+1)$, $(i+1, j)$, $(i+2, j+1)$, and $(i+1, j+2)$. The rectangles A_i, B_i, C_i, and E_i have one corner in each of the columns i to $i + 3$, in the rows 3412, 3142, 2413, and 2143 from left to right (Figure 2).

Lemma 16 (tilted rectangles on narrow grids)**.** *The tilted rectangles of the $3 \times m$ grid are the diamonds $D(i, 1)$. Those of the $4 \times m$ grid are $D(i, 1)$, $D(i, 2)$, A_i, B_i, C_i, and E_i.*

Proof. Each listed set is a rectangle. On three rows, $b = d = 1$, hence $a = c = 1$ and the leftmost corner ℓ is in row 2. On four rows, $b = d = 1$ again gives diamonds. The only other possibility is $\{b, d\} = \{1, 2\} = \{a, c\}$, with ℓ in row $1 + b$. The four choices of sides, $(2, -2), (1, 1)$, $(1, -2), (2, 1)$, $(2, -1), (1, 2)$, and $(1, -1), (2, 2)$, give A, B, C, and E. □

Lemma 17 (boundary cross)**.** *The first row and the first column of the $h \times m$ grid form a valid blackout, so $f(h, m) \geq h + m - 1$.*

Proof. We need only check rectangles showing at most two points (Lemma 4). An axis-aligned rectangle $\{x_1, x_2\} \times \{y_1, y_2\}$ shows $X \times Y$, where $X = \{x_1, x_2\} \setminus \{1\}$ and $Y = \{y_1, y_2\} \setminus \{1\}$. We recover its columns and rows because $|X| = 1$ precisely when $1 \in \{x_1, x_2\}$, and likewise for Y.

By Lemma 13, a tilted rectangle can have only two black corners, ℓ and $(x + a, r - b)$. When both are black, $\ell = (1, 1 + b)$ and the shown corners are $p = (1 + c, 1 + b + d)$ and $q = (1 + a + c, 1 + d)$. Their coordinates determine a, b, c, d. They also differ in both coordinates, so an axis-aligned rectangle could show this pair only if its other two corners were black, which they are not. □

Lemma 18 (axis-aligned rivals)**.** *Let no axis-aligned rectangle be fully black. A* rival *of a set P of at most two white points is an axis-aligned rectangle that shows exactly P. The empty set has no rival, and P has one exactly when it is a singleton $\{(x, r)\}$ with $\{r, s\} \in F$ for some $s \in B_x$, a vertical pair $\{(x, r), (x, s)\}$ with $\{r, s\} \in F$, a horizontal pair $\{(x, r), (z, r)\}$ with $B_x \cap B_z \neq \emptyset$, or an oblique pair $\{(x, r), (z, s)\}$ with $x \neq z$, $r \neq s$, and $(x, s), (z, r) \in S$.*

Proof. A rival of $P \neq \emptyset$ has rows $\{r, s\}$ and columns $\{x, z\}$, with every corner outside P black. Reading those corners gives the listed conditions, which force $z \neq x$ and $s \neq r$ since (x, r) is white. □

5 Three rows

Take $m \geq 3$ with rows $1, 2, 3$ downward. The row bound and boundary cross give $f(3, m) = m+2$, also obtained by the Omega Institute [35]. We now count the optima. The only tilted rectangles to check are the diamonds $D(i, 1)$ (Lemma 16). Denote the columns by $a = \{1\}$, $b = \{2\}$, $c = \{3\}$, $A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{2, 3\}$, and $X = \{1, 2, 3\}$, with a and c *outer*. A blackout without empty columns is a *word* $B_1 \cdots B_m$. We also use each double letter for its pair in F (Section 2).

Proposition 19 (the two types)**.** *Let S be a blackout of the $3 \times m$ grid with $m + 2$ points.*

(a) If S is valid, its word is one X among singletons (type I) or two different doubles among singletons (type II).

(b) For every word of type I or II, G_S is a tree, so axis-aligned rectangles have distinct presentations.

(c) There are $m3^{m-1} + \binom{m}{2} \cdot 6 \cdot 3^{m-2} = m^2 3^{m-1}$ words of type I or II.

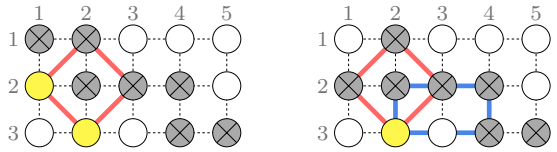
Proof. (a) Lemmas 9 and 10 give $|S| = t + e(H)$, with $t \leq m$ and $e(H) \leq 2$. Equality forces nonempty columns and $\sum_i (|B_i| - 1) = 2$, hence one triple or two doubles. Equal doubles would give a four-cycle in G_S. (b) The nonsingleton columns connect the three rows, and each singleton adds a leaf. (c) Place the triple or ordered pair of distinct doubles, then choose each remaining singleton. □

Classify these *candidates* by their nonsingleton letters, read left to right, as X, AC, CA, AB, BA, BC, and CB. Write AC_m for the valid width-m words in AC, and likewise for the other classes.

Lemma 20 (diamond test)**.** *A candidate S with pair set F is valid if and only if every diamond $D(i, 1)$ passes the test below, where $P = D(i, 1) \setminus S$.*

(a) If $|P| \geq 3$ or P is empty, the diamond passes.

(b) If $P = \{(i + 1, 1), (i + 1, 3)\}$, the diamond fails if and only if $B \in F$.



(a) $aAbCc$ is valid (b) $bAbCc$ fails

Figure 3: The diamond test. (a) In $aAbCc$ the diamond $D(1,1)$ shows the oblique pair $(1,2),(2,3)$, which has no rival since $(1,3)$ is white. (b) In $bAbCc$ it shows only $(2,3)$, and so does the axis-aligned rectangle on columns $2,4$ and rows $2,3$, since $2 \in B_2$ and $C \in F$ (Lemma 20(e)).

(c) If $P = \{(i,2),(i+2,2)\}$, it fails if and only if $B_i \cap B_{i+2}$ meets $\{1,3\}$.

(d) If P is an oblique pair $\{(x,r),(z,s)\}$, it fails if and only if $(x,s),(z,r) \in S$.

(e) If $P = \{(x,r)\}$, it fails if and only if $\{r,s\} \in F$ for some $s \in B_x$.

Proof. Proposition 19(b) distinguishes axis-aligned rectangles. Any rectangle showing at least three points is determined by Lemma 4. An empty diamond has no axis-aligned rival since G_S has no four-cycle. Lemma 18 gives the remaining tests, with $B \in F$ in (b) and $\{r,s\} \in F$ in (e). Two diamonds share at most a tip. For equal presentations, both middle columns would need black rows 1 and 3, impossible in a candidate. □

Figure 3 shows one passing and one failing diamond.

A *window* is three consecutive letters, in columns i to $i+2$. It *passes* or *fails* with $D(i,1)$. Its *tips* $(i,2)$ and $(i+2,2)$ are black exactly when their letters are b, A, C, or X.

Lemma 21 (singleton dichotomy)**.** *In a candidate, a window of three singletons fails if and only if $B \in F$ and it is bab, bbb, or bcb.*

Proof. With two black tips, the vertical pair or one middle-column corner remains. Tests (b) and (e) reject exactly when $B \in F$. With one black tip, three corners remain or test (d) passes because $(i+1,2)$ is white. With neither, at least three corners remain. □

Lemma 22 (six of nine in the class $\{A,C\}$)**.** *Let S be a candidate with $F = \{A,C\}$, and let s,t be singletons.*

(a) A window sAt fails if and only if st is bb, bc, or cb, also for $t = C$ read as $t = b$.

(b) A window sCt fails if and only if st is bb, ba, or ab.

(c) A window with one double, at a tip, passes.

Proof. (a) Only $(i+1,3)$ is white in the middle column. The right tip $(i+2,2)$ is black for $t \in \{b,C\}$. If both tips are black, test (e) fails because $2 \in B_{i+1}$ and $C \in F$. With just the right tip black, the pair $(i,2),(i+1,3)$ fails test (d) exactly for $s = c$. The other tip is

symmetric. With neither black, three corners show. (b) Reflect across the middle row. (c) Apply Lemma 21 to *Ast* with the left tip already black. No case fails since $B \notin F$. Reflection and reversal give the remaining positions. □

Write a word in AC_m as $UAVCW$, with singleton strings U, V, and W. The diamond test reduces validity to checking each window. Set $G_{AC}(z) = \sum_{m\geq 2} |AC_m| z^m$. Define G_{AB} by counting $UAVBW$, and $G_X(z) = \sum_{m\geq 1} |X_m| z^m$ by counting UXW. Write $\Phi = 1 - 3z$. A nonempty singleton string with a specified letter next to a double contributes z/Φ.

Proposition 23 (the class $\{A, C\}$)**.** *The generating function of AC is*

$$G_{AC}(z) = \frac{z^2 - 4z^3 + 10z^4 - 8z^5 - 3z^6}{(1-3z)^3}, \tag{2}$$

and

$$|AC_m| = 2(m-1)(3m+2)3^{m-5} \qquad (m \geq 4). \tag{3}$$

Proof. Lemma 22(a) and the half turn allow tAC only for $t = a$ and ACt only for $t = c$. Each AtC passes tests (b) and (e). Lemmas 21 and 22(c) leave only windows centered on a double.

We count by the length of the gap V. With no gap, a nonempty U must end in a and a nonempty W must start in c, contributing $z^2(1 + z/\Phi)^2$. For $V = t$, the allowed last letters of U number $p_t = 3, 1, 2$, and the allowed first letters of W number $q_t = 2, 1, 3$, for $t = a, b, c$. Thus $\sum p_t = \sum q_t = 6$ and $\sum p_t q_t = 13$, giving $z^3 \sum_t (1 + p_t z/\Phi)(1 + q_t z/\Phi) = z^3(3 + 12z/\Phi + 13z^2/\Phi^2)$. For $|V| \geq 2$, each end contributes $3 + 6z/\Phi$ (a lone endpoint letter, or one of six passing pairs). The unrestricted middle contributes $1/\Phi$, so this case gives $9z^4(1-z)^2/\Phi^3$. Adding the three cases gives (2). The identity $[z^m] z^k \Phi^{-3} = \binom{m-k+2}{2} 3^{m-k}$ then gives (3). □

Theorem 24 (the number of three-row optima)**.** *Let E_m count the optima of the $3 \times m$ grid with a column black in rows 1 and 3 (A398448). For $m \geq 4$,*

$$a_3(m) = 4(m-1)(3m+2)3^{m-5} + E_m. \tag{4}$$

Proof. A column black in rows 1 and 3 must be B or X. Proposition 19(a) therefore identifies these optima with the classes other than AC and CA. Reversing columns pairs AC, AB, and BC with CA, BA, and CB, respectively. Reflection across the middle row also pairs AB with CB. Hence

$$a_3(m) = 2|AC_m| + 4|AB_m| + |X_m| = 2|AC_m| + E_m \qquad (m \geq 3), \tag{5}$$

and substituting (3) gives (4). □

5.1 The generating function

Put $L = 1 - 2z - 2z^2$, $N = 1 + 2z^2$, and $\Psi = LN$.

Theorem 25 (three-row generating function, parts (b) and (c) computer-assisted)**.**

(a) The series $\sum_{m\geq 3} a_3(m)z^m$ *equals* $2G_{AC} + 4G_{AB} + G_X - z - 12z^2$*, with* G_{AB} *and* G_X *from Propositions 29 and 28, of denominators dividing* Ψ^3 *and* Ψ^2*.*

(b) Its reduced denominator is exactly $Q = \Phi^3 L^3 N^3$*.*

(c) With

$$\begin{aligned} P(z) = z &- 3z^2 - 34z^3 + 228z^4 - 641z^5 + 1307z^6 - 2186z^7 \\ &+ 2056z^8 - 236z^9 - 1060z^{10} + 2584z^{11} - 4264z^{12} \\ &- 1440z^{13} + 704z^{14} - 1024z^{15} + 2080z^{16} + 2176z^{17} + 384z^{18}, \end{aligned}$$

we have

$$\sum_{m\geq 3} a_3(m)z^m = \frac{P(z)}{Q(z)} - z - 12z^2. \tag{6}$$

Proof. For (a), sum (5), removing widths one and two by subtracting $z + 12z^2$. Part (b) is proved in Section 5.2. For (c), exact algebra in SymPy [27] gives P. The script `three_row_closed_form_check.py` independently checks (6) against a transfer matrix through width 100. □

Let $\rho = 1 + \sqrt{3}$ and $\sigma = 1 - \sqrt{3}$.

Theorem 26 (three-row closed form, computer-assisted exact algebra)**.** *For every* $m \geq 3$*,*

$$E_m = p_+(m)\rho^m + p_-(m)\sigma^m + \eta_m, \tag{7}$$

where $432\,p_\pm(m) = (19 \pm 11\sqrt{3})m^2 + (61 \pm 41\sqrt{3})m + 104 \pm 28\sqrt{3}$ *and* $\eta_m = (-2)^{\lfloor m/2 \rfloor}\,\kappa(m)$ *with* $\kappa(m) = (2m^2 + 41m - 104)/216$ *for even* m *and* $\kappa(m) = (7m^2 - 32m - 175)/432$ *for odd* m*. Hence* $E_m \sim \frac{19+11\sqrt{3}}{432} m^2 \rho^m$*. The hand proof gives* $E_m > 0$ *and*

$$a_3(m) = \frac{4}{81} m^2 3^m - \frac{4}{243}(m+2)3^m + O(m^2\rho^m). \tag{8}$$

Proof. By (5), E_m is the coefficient of z^m in $4G_{AB} + G_X$, whose denominator divides L^3N^3. The possible poles are $1/\rho$, $1/\sigma$, and $\pm 1/(i\sqrt{2})$, each of order at most three. Apart from a polynomial part of degree one, their coefficients have the form (7). Exact algebra determines the twelve rational constants. Both sides obey the order-12 recurrence from L^3N^3 for $m = 14$ onward and agree at $m = 2, \ldots, 13$. The leading term follows from $|\sigma| < 1$ and $|\eta_m| = O(m^2 2^{m/2})$. For the hand bound, the pole $1/\rho$ of least modulus gives $E_m = O(m^2\rho^m)$, which yields (8) through (4). Finally, the boundary cross has word $Xa\cdots a$ (Lemma 17), so $E_m > 0$. □

5.2 The classes AB and X

We count X and AB by recording the last two singleton letters. We then rule out cancellation in Theorem 25(b). Keep $\Psi = LN = 1 - 2z - 4z^3 - 4z^4$. Call a singleton string *restricted* if no two b letters are at distance two. Reversal and reflection carry AB to BA, BC, and CB.

Lemma 27 (forbidden windows in the classes containing B)**.** *In classes AB and X, the table lists exactly the failing windows. Here s is any singleton.*

class	*windows*	*forbidden*
AB, X	*singletons only*	*bsb*
AB	*one double*	*bsA, Asb, bAb, bAc, cAb, aBa, aBb, bBa, cBc*
	two doubles	*bAB, ABa*
X	*one triple*	*bsX, Xsb, aXa, aXb, bXa, bXc, cXb, cXc*

Proof. Lemma 21 handles singleton windows. For sAt, use Lemma 22(a), also rejecting bAb by test (e) since $B \in F$. In sBt or sXt, the middle corners are black. For $s = t = b$ the diamond is empty. For outer letters, test (c) fails exactly for $s = t$. For $s = b \neq t$, only $(i+2, 2)$ shows, and test (e) fails exactly when $\{2, t\} \in F$. In uAB, the black corners $(i+2, 3), (i+1, 2)$ make test (d) fail for $u = b$, while otherwise three corners show. In ABw, $w = b$ gives an empty diamond, and test (e) rejects $w = a$. Tests (b) and (e) exclude Asb and Xsb. Other left-tip windows show three corners or need the white point $(i+1, 2)$ for an oblique rival. Reflect for the other cases. □

Encode outer letters by bit 0 with weight 2, and b by bit 1. Let h_{ij} count continuations after bits i, j, forbidding b after $1, j$. Read each run $w_1 \cdots w_n$, $n \geq 1$, away from its double. Let g_0, g_1 count runs with a prescribed outer first letter or $w_1 = b$, respectively. Let u, v count the same runs with w_2 outer if present. A fictitious preceding letter, outer for g and b for u, v, gives $(g_0, g_1, u, v) = z(h_{00}, h_{01}, h_{10}, h_{11})$. Appending an allowed letter gives the recurrence, whose solution is

$$\Psi g_0 = z(1 + z + 3z^2 + 2z^3), \quad \Psi g_1 = z(1 + z), \quad \Psi u = z(1 + 2z^2 + 2z^3), \quad \Psi v = z. \tag{9}$$

Proposition 28 (the triple class)**.** *With u and v as in (9), $G_X(z) = z\big(1+4u+2v+2u^2+v^2\big)$, and its denominator divides Ψ^2.*

Proof. In UXW, both runs are restricted. Forbidding bsX and Xsb makes the second letter from X outer. If just one side is nonempty, it contributes $2u + v$ on either side. With both nonempty, the allowed neighbor pairs are (a, c), (b, b), and (c, a), contributing $2u^2 + v^2$. □

In $UAVBW$, all singleton runs are restricted. Reading outward, U from A uses u or v, and W from B uses g_0 or g_1 (Lemma 27). Let ℓ_t and r_t count these runs, including the empty run, when t follows A or precedes B, respectively. The windows centered on doubles give

$$(\ell_a, \ell_b, \ell_c) = (1 + 2u + v, 1 + u, 1 + 2u), \ (r_a, r_b, r_c) = (1 + g_0, 1 + g_0 + g_1, 1 + g_0 + g_1). \tag{10}$$

To join the runs, we must also record the final bit. For $s \in \{00, 10\}$, let $J_{s\beta}$ count continuations from s ending in β, assigning the empty continuation to $\beta = 0$. The same recurrence gives the matrix below, with rows indexed by s and columns by β

$$J = \frac{1}{\Psi}\begin{pmatrix} 1 + 2z^2 + 2z^3 & z + z^2 \\ 1 & 2z^2 + 2z^3 \end{pmatrix}. \tag{11}$$

Proposition 29 (the class $\{A, B\}$, polynomial form computer-assisted)**.** *With* (9) *to* (11)*,*

$$\begin{aligned} G_{AB}(z) = z^2(1+2u)(1+g_0+g_1) + z^3(\ell_a r_a + \ell_b r_b + \ell_c r_c) \\ + z^4\,(\ell_a + \ell_c,\ \ell_b)\,J\,(r_a + r_c,\ 2r_b)^T. \end{aligned} \tag{12}$$

Moreover $G_{AB} = P_B/\Psi^3$ *with*

$$P_B(z) = z^2(1+z)(1+z^2-3z^3+4z^6+4z^7). \tag{13}$$

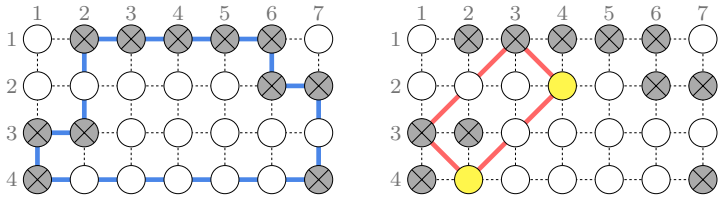
Proof. Let k be the length of V. For $k = 0$, the boundary windows uAB and ABw require $u \neq b$ and $w \neq a$, giving the first term. For $k = 1$, AtB always passes, while uAt and tBw contribute $\ell_t r_t$.

For $k \geq 2$, Av_1v_2 forces v_2 to be outer, and $v_{k-1}v_kB$ always passes. The remaining string $v_3 \cdots v_k$ continues from bits 00 or 10, according as v_1 is outer or b, with boundary weight $\ell_{v_1} r_{v_k}$. The entry $r_a + r_c$ counts the two outer choices for v_k. For $k \geq 3$, the outer-letter weight 2 in J and the factor 2 in $2r_b$ account for the two choices of v_2. The constant term of J handles $k = 2$. This gives (12), and exact algebra gives (13). □

Proof of Theorem 25(b). The denominator divides $\Phi^3\Psi^3 = Q$ by part (a). Its factors share no roots, since $\Psi(\frac{1}{3}) \neq 0$ and $L - N = -2z - 4z^2$ vanishes only where $N \neq 0$. At $z = \frac{1}{3}$, G_{AC} alone has a triple pole, its numerator being $\frac{4}{81}$. At roots of Ψ, the remainders $P_B \bmod L = \frac{z}{2}$ and $P_B \bmod N = \frac{1}{4} - \frac{z}{2}$ are nonzero. Thus G_{AB} has triple poles there, which neither the double poles of G_X nor the regular G_{AC} can cancel. □

Remark 30 (optima up to symmetry). Counting up to row reflection and column reversal gives $o_3(m)$, with first values 15, 58, 227, 864, 3190, and 11502 for widths 3 to 8 (A401226). Burnside's lemma says that $4o_3(m)$ equals $a_3(m)$ plus the fixed counts of the three nonidentity symmetries. Row reflection allows only b, B, and X. An optimum must therefore have one X and otherwise b. For $m \geq 4$, one of Xbb, bbX, or bbb occurs and fails (Lemmas 20(b) and 21). Thus only bXb is fixed.

For the half turn, put $m = 2k$ or $m = 2k+1$. It reverses the word and exchanges A with C and a with c. A fixed two-double word has A and C in mirror columns $j \leq k$ and $m+1-j$, with b at the center for odd m. The first k letters determine the rest. Only the windows centered on A and C can fail (the proof of Proposition 23), and these are mirror images. Lemma 22(a) leaves 3^{k-1} words in AC when $j = 1$, 3^{k-2} when $j = k \geq 2$, and $2 \cdot 3^{k-2}$ when $1 < j < k$. The count in CA is equal. Summing gives $4k3^{k-2}$ for $k \geq 2$. At $m = 3$ there are just AbC and CbA. Their combined series is $W = 2z^3 + \frac{4}{9}(1+z)\big(3z^2(1-3z^2)^{-2} - 3z^2\big)$.



(a) the cycle of G_S (b) A_1 shows two points

Figure 4: The blackout (14) for $m = 7$. (a) Its row-column graph is an eight-cycle, traced through the black points, with leaves. (b) The weak rectangle A_1 shows $(2, 4)$ and $(4, 2)$, the case $i = 1$ of the table in the proof of Theorem 31.

Column reversal forces a palindrome. Every noncentral letter is repeated, so a fixed optimum must have odd width and form UXU^{rev}. The middle window tXt requires $t = b$, and the forbidden windows bsX and Xsb force the next letter outward to be outer (Lemma 27). Thus U is counted by v in (9). For a half-turn-fixed triple word $UX\iota(U^{\mathrm{rev}})$, where ι reflects rows, the allowed middle pairs are (a, c), (b, b), and (c, a), giving $2u + v$ for U. Adding z^3 for bXb, $2z\big(u(z^2) + v(z^2)\big)$ for the paired runs, and W for two doubles gives

$$4\sum_{m\ge 3} o_3(m)z^m = \sum_{m\ge 3} a_3(m)z^m + z^3 + \frac{4z^3(1+z^4+z^6)}{M(1+2z^4)} + \frac{2z^3(1-z)(1+5z+3z^2-3z^3)}{(1-3z^2)^2},$$

Here $M = 1-2z^2-2z^4$, so $\Psi(z^2) = M(1+2z^4)$. The independent check `three_row_orbits.py` agrees with the fixed counts through width 24 and the orbit counts $o_3(m)$ for $m \le 10$.

6 Four rows

Theorem 31 (four rows)**.** *For every $m \ge 6$ we have $f(4, m) = m + 4$.*

We attain the row bound by placing the extra black points near the two ends. Take

$$S = \{(i, 1) : 2 \le i \le m-1\} \cup \{(m-1, 2), (m, 2), (1, 3), (2, 3), (1, 4), (m, 4)\}. \tag{14}$$

Here $|S| = m+4$ and G_S is the eight-cycle $r_1\, c_2\, r_3\, c_1\, r_4\, c_m\, r_2\, c_{m-1}$ with leaves $c_3, \dots, c_{m-2}$ on r_1 (Figure 4).

Proof of Theorem 31. The graph G_S has no four- or six-cycle. Thus Lemma 7 distinguishes the axis-aligned rectangles, and none is fully hidden.

For tilted rectangles, use Lemma 16. Their corners in row 1 are black and those in row 4 are white. For $D(i, 1)$ the additional black corners are $(i+2, 2)$ when $i \ge m-3$ and $(i+1, 3)$ when $i = 1$. For $D(i, 2)$ they are $(i, 3)$ when $i \le 2$ and $(i+1, 2)$ when $i = m-2 \ge 4$. For A_i and B_i they are $(i, 3)$ when $i \le 2$ and $(i+3, 2)$ when $i \ge m-4$. It remains to check the weak rectangles in the table. Here $s = 1$ for A_i and $s = 2$ for B_i.

rectangle	which i	corners shown
$D(1,1)$		$(1,2),(3,2)$
$D(i,1)$	$i \in \{m-3, m-2\}$	$(i,2),(i+1,3)$
A_i or B_i	$i \in \{1,2\}$, $i \notin \{m-4, m-3\}$	$(i+s,4),(i+3,2)$
A_i or B_i	$i \in \{m-4, m-3\}$, $i \notin \{1,2\}$	$(i,3),(i+s,4)$
A_i or B_i	$i = 2$, $m = 6$	$(i+s,4)$

The first four cases have different shown row sets. Within any one, the point in row 2 or 3 determines i, and the row-4 point determines s. We now rule out axis-aligned rivals. No row r has both $(1,r)$ and $(3,r)$ black, so $\{(1,2),(3,2)\}$ has no rival. In the next three cases, the axis-aligned rectangle through the pair has a white corner, respectively $(i,3)$, $(i+s,2)$, or $(i,4)$. Here we use $i \geq 3$, $i+s \leq 4 < m-1$, and $3 \leq i \leq m-3$, respectively. The final case has $m = 6$ and one shown point $(z,4)$ with $z \in \{3,4\}$. Column z is black only in row 1, and no column is black in rows 1 and 4, so this singleton also has no rival. □

For $m \geq 4$, Theorem 11 leaves $3\,(m)_4\,4^{m-4}$ *candidates* of size $m+4$. Each has four doubles forming a four-cycle F on the rows, with a singleton in every remaining column.

Corollary 32 (widths four and five, computer-assisted)**.** *We have* $f(4,4) = 7$ *and* $f(4,5) = 8$.

Proof. The boundary cross gives the lower bounds. For the upper bounds, the search finds an ambiguous pair in each of the 72 and 1,440 candidates, respectively. □

Theorem 33 (no hidden rectangle)**.** *For* $m \geq 4$*, no valid blackout of the* $4 \times m$ *grid hides a rectangle.*

Proof. If a rectangle is hidden, we show that two others have the same presentation. By Lemma 2, it suffices to write its corner set as $C(R_1) \triangle C(R_2)$ for $R_1 \neq R_2$. For an axis-aligned rectangle, use its two rows and pair each column x, y with a third column w. For $1 \leq i \leq m-3$, put $\mathsf{Q}_i = \{(i,2),(i,3),(i+3,2),(i+3,3)\}$. The tilted cases follow from

$$\begin{aligned} \mathsf{C}_i &= \mathsf{A}_i \triangle \mathsf{Q}_i, & \mathsf{E}_i &= \mathsf{B}_i \triangle \mathsf{Q}_i, \\ D(i+1,1) &= D(i,2) \triangle \mathsf{A}_i, & D(i+1,2) &= D(i,1) \triangle \mathsf{E}_i, \end{aligned} \tag{15}$$

since any of the three sets in an identity is the difference of the other two. The first line handles A_i, B_i, C_i, and E_i. For a diamond $D(j,1)$ or $D(j,2)$, use the second line with $i = j-1$ or $i = j$. One choice is always available, since $j \geq 2$ or $j \leq m-3$ when $m \geq 4$. □

Remark 34 (forbidden-block bound). For every $m \geq 6$ we have $a_4(m) \leq 3\,(m)_4\,4^{m-4}(53/64)^k$, where $k = \max(0, \lfloor m/4 \rfloor - 4)$. So $a_4(m) = O\big(m^4\,212^{m/4}\big)$. It is weaker than Corollary 38, but its proof is by hand.

Proof. Choose F, the four double columns, and their pairs in $3\,(m)_4$ ways. Call four consecutive singleton columns a *block*, recorded by their black rows. Starting at column x, the

four *template words* 2413, 3412, 2143, and 3142 are the corner rows of C_x, A_x, E_x, and B_x. Fix one template rectangle τ with word $t_1t_2t_3t_4$.

Swapping t_p and t_q gives a *swap word*. It cannot occur, since τ and the axis-aligned rectangle through $(x+p-1, t_p)$ and $(x+q-1, t_q)$ would show the same pair. The templates are exactly the permutations whose end letters both lie in $\{2,3\}$. Swaps within the end or middle positions give templates again. Swaps between these positions give the sixteen permutations with one end letter in $\{2,3\}$. Together these are 20 forbidden swap words.

An *edit word* changes one letter $t_p = r$ to s with $\{r,s\} \in F$. This is also impossible. The rectangle τ and the axis-aligned rectangle on rows r, s and columns $x+p-1$ and d, where d is the double column for $\{r,s\}$, would show only $(x+p-1, r)$. There are 8 edits per template. Two templates can share an edit only when two positions contain r, s in opposite orders. They then share two edits if $\{r,s\} \in F$. This happens for two template pairs with $\{r,s\} = \{2,3\}$ and two with $\{1,4\}$. Hence at least $32 - 8 = 24$ edit words are forbidden. Each repeats a letter, so none is a swap word. A block therefore allows at most $256 - 20 - 24 = 212 = 4^4 \cdot 53/64$ words.

Partition the columns into the $\lfloor m/4 \rfloor$ groups $4j-3, \ldots, 4j$. At most four contain a double, leaving at least k singleton blocks. All other singleton columns have at most 4 choices, which gives the bound. □

Lemma 35 (six columns of memory)**.** *Let $m \geq 6$, and let F be the edge set of a four-cycle on the rows. A word over the four singleton letters and the pairs of F, each pair used once, represents an optimum if and only if (i) no tilted rectangle is hidden, (ii) no tilted rectangle shares its presentation with an axis-aligned rectangle whose columns are among its own and the double columns, and (iii) no two distinct tilted rectangles in a block of at most seven consecutive columns share a presentation.*

Proof. Necessity follows from validity and Theorem 33. Conversely, the prescribed doubles and singletons give an axis-valid word by Theorem 11. No axis-aligned rectangle is hidden, since that would give a four-cycle in G_S, and (i) excludes hidden tilted rectangles.

If two rectangles nevertheless show the same P, then $P \neq \emptyset$ and at least one is tilted. If the other is axis-aligned, a column of it either meets P or has two black corners. In the first case it is a column of the tilted rectangle, and in the second it is a double column. Thus (ii) excludes the pair. If both are tilted, choose a common column c meeting P. Each rectangle spans at most four columns, so their union fits in $c-3$ to $c+3$, contradicting (iii). The word is valid and has $m+4$ black points, hence is an optimum. □

We use this memory bound to count optima. In a word of singletons and pairs of F, a tilted rectangle *fails locally* if it is hidden, has a rival under Lemma 18 with pair set F, or shares its presentation with another tilted rectangle.

Proposition 36 (the automaton)**.** *Let F be the edge set of a four-cycle on the rows. The automaton $\mathcal{A}_F$ reads the four singleton letters and the pairs in F. It starts at $(\epsilon, \emptyset)$, where ϵ is the empty word. A state (u, U) records the suffix of at most six letters read so far and the set $U \subseteq F$ of pairs already used.*

On reading y, it rejects if $y \in U$ or if a tilted rectangle in uy ending in the new column fails locally. Otherwise it replaces u by the suffix of uy of length $\min(6, |uy|)$ and U by $U \cup (\{y\} \cap F)$. A word is accepted if it is read to the end without rejection and the final register is $U = F$. For each $m \geq 6$, the accepted words of length m are exactly the optima of the $4 \times m$ grid with pair set F.

Proof. Take $m \geq 6$. At its last column, each tilted rectangle lies in uy, since its span is at most three. An accepted word uses every pair of F once, so it is a candidate. Presentations of at least three points are already unique (Lemma 4). The local tests rule out hidden tilted rectangles and axis-aligned rivals, giving (i) and (ii) of Lemma 35. For (iii), any two tilted rectangles within seven consecutive columns fit in memory when the later last column is read. Their presentations are compared then. Lemma 35 makes the accepted word an optimum.

Conversely, an optimum with pair set F uses every pair once. Theorem 33 excludes hidden rectangles, and validity excludes every repeated presentation. Applied with F, Lemma 18 cannot find a rival. The word passes each step and finishes with register F, so it is accepted. □

The script `four_row_equivalence_check.py` builds $\mathcal{A}_F$ (60,975, 138,095, and 55,575 states) and checks equivalence with a smaller machine from an earlier search. Including the pair register, that machine has 2,519, 3,089, and 2,119 reachable states. Two transfer counters agree for $m \leq 1500$, and direct candidate tests agree for $m \leq 10$ (Table 1).

m	6	7	8	9	10	11	12
$\{1,2\} \mid \{3,4\}$	0	4	128	920	4,852	24,112	114,424
$\{1,3\} \mid \{2,4\}$	20	256	1,780	11,128	63,540	346,320	1,785,768
$\{1,4\} \mid \{2,3\}$	6	32	210	1,284	7,006	32,224	137,656
$a_4(m)$	26	292	2,118	13,332	75,398	402,656	2,037,848

Table 1: Four-row optima by pair set (computer-assisted).

Theorem 37 (the four-row generating function, computer-assisted)**.** *There are coprime integer polynomials $P_4 = 26z^6 - 826z^7 + \cdots$ of degree 444 and $Q_4 = \sum_{j=0}^{420} q_j z^j$ of degree 420 with $q_0 = 1$ and $\sum_{m \geq 6} a_4(m) z^m = P_4/Q_4$. The recurrence $\sum_j q_j\, a_4(n-j) = 0$ holds for every $n \geq 445$ and fails at $n = 444$. Over the integers*

$$Q_4 = (1-z)^2(1+z)^2\, g_5^3\, \tilde{g}_5^3\, g_6^3\, g_{11}^5\, g_{17}^3\, g_{19}^3\, g_{20}^5\, g_{21}^5,$$

where g_d and $\tilde{g}_d$ are irreducible of degree d with constant term 1.

Proof. Proposition 36 identifies the counted words with the optima. Running Berlekamp–Massey modulo eight primes on $a_4(m)$ for $m \leq 1500$ produces P_4 and Q_4. To check all widths, let M_F be the transfer matrix on reachable states and write $v(n) = v(0)M_F^n$. The state vectors satisfy $\sum_j q_j v(n_F - j) = 0$ at $n_F = 436$, 435, and 445 for the three machines. Multiplying by powers of M_F proves the recurrence for all $n \geq 445$. Since $\deg P_4 = 444$, it fails at $n = 444$. The displayed factorization is obtained in SymPy. □

Corollary 38 (growth, computer-assisted)**.** *Let $\rho_4 = 3.351996777310\ldots$ be the largest real root of $p(x) = x^{20}g_{20}(1/x) = x^{20} - 4x^{19} + 2x^{18} + \cdots - 26x^2 - 4x + 8$. There is a polynomial π of degree 4 with $a_4(m) = \rho_4^m \pi(m) + O((1000/341)^m)$, so $a_4(m) \sim c\, m^4 \rho_4^m$ with $c \approx 8.1611 \times 10^{-5}$.*

Proof. Isolate the roots of every factor of Q_4 exactly. The only root of modulus at most 0.341 is $r = 1/\rho_4 = 0.29832\ldots$, simple in g_{20} and therefore of order five in Q_4. Separate its principal part as $P_4/Q_4 = \sum_{j=1}^{5} c_j(1 - z/r)^{-j} + E(z)$. The coefficients of E are $O((1000/341)^m)$. Coprimality gives $c_5 = H(r) \neq 0$, where $H(z) = P_4(z)(1 - z/r)^5/Q_4(z)$. Now $[z^m](1 - z/r)^{-j} = \rho_4^m \binom{m+j-1}{j-1}$, so $\pi(m) = \sum_j c_j \binom{m+j-1}{j-1}$ and $c = c_5/24$. Interval arithmetic encloses c between $8.1610970775 \times 10^{-5}$ and $8.1610970776 \times 10^{-5}$. □

7 Five, six, and seven rows

7.1 Boundary padding

We extend a finite construction by adding black points along a boundary row. The next theorem reduces validity at every width to one finite check. For a blackout S of the $h \times m$ grid and $\lambda, \mu \geq 0$, form $S_{\lambda,\mu}$ by adding columns $1 - \lambda, \ldots, 0$ and $m + 1, \ldots, m + \mu$, black only in a chosen *padding row*, row 1 or row h.

Theorem 39 (finite collar)**.** *Let $h \geq 3$, fix the padding row and the padded sides, and let λ^* and μ^* be W_h of Lemma 14 on padded sides and 0 otherwise. Then $S_{\lambda,\mu}$ is valid for all $\lambda, \mu \geq 0$ vanishing on unpadded sides if and only if S_{λ^*,μ^*} is valid.*

Proof. Set $S^* = S_{\lambda^*,\mu^*}$. Necessity follows by choosing these padding lengths. For sufficiency, suppose S^* is valid and put $U = S_{\lambda,\mu}$.

A tilted rectangle has at most one corner in row 1 and at most one in row h (Lemma 13 and (1)). A weak tilted rectangle of U therefore has a black corner in an original column e. Its span is bounded by Lemma 14, so its columns lie in $[e - W_h, e + W_h] \cap [1 - \lambda, m + \mu] \subseteq [1 - \lambda^*, m + \mu^*]$. It lies in S^* and has the same presentation there.

Suppose distinct R_1, R_2 in U show the same P.

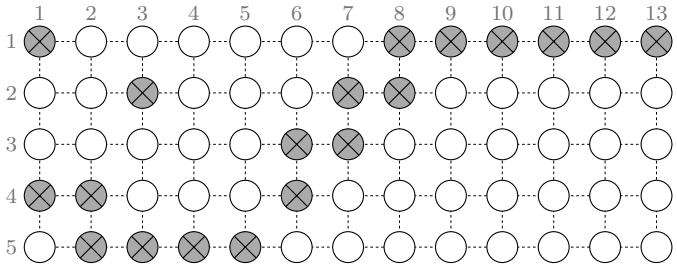
Both axis-aligned. Every added column has degree one in G_U, so any cycle already lies in G_{S^*}. Since S^* is valid and has at least three columns, Lemma 7 makes U axis-valid.

Both tilted. Lemma 4 gives $|P| \leq 2$. Both rectangles are weak, so both lie in S^* with presentation P, contradicting its validity.

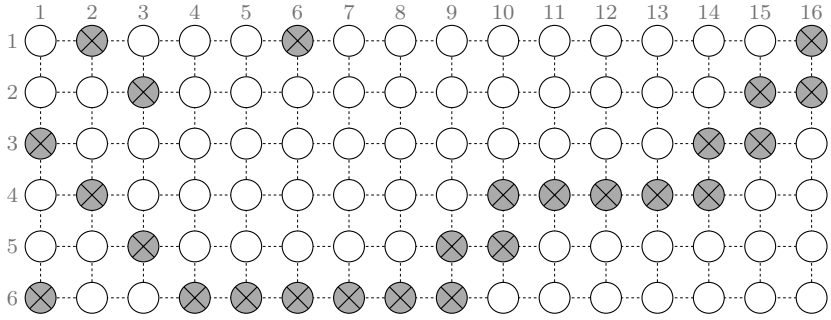
One of each. Take R_1 tilted, hence already in S^*. A column of R_2 meeting P also belongs to R_1. A column not meeting P has two black corners, so it must be original. Thus R_2 also lies in S^* with the same presentation, again a contradiction. □

7.2 Five rows

List each column's black rows from left to right. On five rows, start with the blackout S_8 given by $14, 45, 25, 5, 5, 34, 23, 12$ of the 5×8 grid, so that $B_1 = \{1, 4\}$ and $B_4 = \{5\}$.



(a)



(b)

Figure 5: (a) The 5×13 test board of S_8. (b) The 6×16 blackout S_{16}.

Theorem 40 (five rows, (a) and (c) computer-assisted, (b) checked in Lean)**.**

(a) We have $f(5, m) = m + 5$ for $m = 5$, 6, and 7.

(b) We have $f(5, m) = m + 6$ for every $m \geq 8$, attained by S_8 followed by $m - 8$ columns black in row 1 only.

(c) The 5×8 grid has exactly four maximum blackouts, the images of S_8 under row reflection and column reversal.

Proof. (b) The upper bound is Theorem 8. For the construction, $W_5 = 5$, so the collar theorem with right padding in row 1 reduces the family to the test board in Figure 5(a). Three enumerators and Lean verify that the 954 rectangles of this test board have different presentations. (a) At $m = 5$, use Corollary 12 and the construction S_0 in Theorem 46. At $m = 6, 7$, the constructions are $15, 12, 23, 3, 34, 45$ and $1, 12, 23, 3, 34, 45, 15$, recorded in `jis/data/small_cells_sat_5x6.json` and `jis/data/small_cells_sat_5x7_b12.json`. They give the lower bounds. Checked SAT certificates rule out $m+6$ points (Section 11). (c) Two SAT enumerations find the four images. A separate column search checks all $10\,(8)_6\,5^2 = 5{,}040{,}000$ equality candidates from Theorem 11 and finds the same four. □

Theorem A gives $f(h, m) \leq h + m$ whenever a side is at most four, and on 5×5, 5×6, 5×7, and 6×6. Up to transposition, these exhaust the grids of area below 40. Thus 5×8 is the first failure of the guess $f(h, m) = h + m$.

Remark 41 (five-row counts, computer-assisted). The counts $a_5(m)$ for $8 \le m \le 23$ are A401225. They begin 4, 44, 350, 3,004, 24,706, and 193,406, and $a_5(23) = 5{,}510{,}071{,}794{,}736$. For $m \ge 13$ each count agrees in two independent transfer calculations using the equality structure of Theorem 11.

7.3 Six rows and small grids

Let S_{16} be the blackout of the 6×16 grid in Figure 5(b). Its doubles, in the columns $1, 2, 3, 6, 9, 10, 14, 15, 16$, carry the edges of $K_{3,3}$ between $\{1, 3, 5\}$ and $\{2, 4, 6\}$.

To exclude blackouts one point below the row bound, we first classify their possible row graphs.

Lemma 42 (one point below the bound)**.** *Let $m \ge 3$, and let S be an axis-valid blackout of the $6 \times m$ grid with $m+8$ points. Then every column of S is empty, a singleton, or a double, the doubles have distinct row pairs, and one of the following holds.*

(a) No column is empty, and the pairs of the doubles are the edges of $K_{3,3}$ minus one edge.

(b) No column is empty, and the pairs of the doubles are the edges of $K_{2,4}$.

(c) Exactly one column is empty, and the pairs of the doubles are the edges of $K_{3,3}$.

Proof. For every choice of trees, Lemma 9 gives $|S| = t + e(H)$, where $t \le m$ counts nonempty columns. Since $e(H) \le 9$ by Lemma 10, $(t, e(H))$ is $(m, 8)$ or $(m-1, 9)$.

In the second case H has q_6 edges. The proof of Theorem 11 makes H a $K_{3,3}$ and excludes columns with three black rows, giving (c).

In the first case H is triangle-free with 8 edges. If it were not bipartite, its shortest odd cycle would be a chordless five-cycle. The sixth vertex could meet at most two cycle vertices, giving $e(H) \le 7$, a contradiction. Hence H is bipartite with parts of sizes k and $6-k$, and $k(6-k) \ge 8$ forces H to be $K_{2,4}$ or $K_{3,3}$ minus an edge. In both graphs, two vertices in the same part have at least two common neighbors. If $|B_i| \ge 3$, then T_i has a vertex v with neighbors u and w. These share a part of H, so they have a common neighbor $z \ne v$. Replace vw in T_i by uw. This gives another tree on B_i but creates the triangle uwz in H, contrary to Lemma 9. Thus every column is a singleton or a double. The double pairs are the edges of the simple graph H, so they are distinct. □

There are $10 \cdot 9 = 90$ labeled graphs in case (a), 15 in case (b), and 10 in case (c).

Theorem 43 (six rows, (a) and (b) computer-assisted, (c) checked in Lean)**.**

(a) For $8 \le m \le 11$ we have $f(6, m) = m + 7$.

(b) For $12 \le m \le 14$ we have $f(6, m) = m + 8$, attained by the blackout $16, 36, 25, 12, 1^{m-8}, 23, 34, 45, 56$, where 1^k denotes k columns black only in row 1.

(c) For every $m \ge 15$ we have $f(6, m) = m + 9$.

So $f(h, m)$ is known for every $h \leq 6$ and every m.

Proof. For $9 \leq m \leq 14$, an optimum with $m+9$ points would have double columns carrying $K_{3,3}$, by Theorem 11. Checked LRAT certificates exclude all ten row partitions in the structured encoding (Lemma 54). At $m = 8$, Corollary 12 gives the same bound. Hence $f(6, m) \leq m + 8$ for $8 \leq m \leq 14$.

(a) The case $m = 8$ is Theorem 45. For the lower bound at $9 \leq m \leq 11$, the blackout $12, 23, 34, 45, 6, 6, 36, 56, 25$ of the 6×9 grid is valid with 16 points. Its collar test board, with $W_6 = 7$ columns black only in row 1 on each side, is valid (`six_row_collar_6x9_check.py`), so Theorem 39 gives $m + 7$ points at every $m \geq 9$. For the upper bound, Lemma 42 leaves 115 structured instances at each width, one for each labeled graph in its three cases. They are unsatisfiable under two encodings, and encoding A has checked LRAT certificates.

(b) Three enumerators check that the blackout is valid. It has $m + 8$ points.

(c) Again the row bound gives the upper bound. For $m = 15$, take $56, 45, 34, 23, 1, 1, 14, 1, 1, 12, 36, 1, 25, 6, 16$ from `jis/data/six_row_sat_w15_A.json`, which is valid with 24 points. For $m \geq 16$, apply the collar theorem to S_{16} in Figure 5(b). Since $W_6 = 7$, it suffices to add $(17, 1), \ldots, (23, 1)$. Three enumerators and Lean verify distinct presentations for all 4,627 rectangles of that test board. □

Theorem 44 (seven rows, computer-assisted)**.** *For every $m \geq 24$ we have $f(7, m) = m+12$.*

Proof. The row bound gives $f(7, m) \leq m + 12$. For the lower bounds, structured encoding A of Section 11 uses twelve doubles on the edges of $K_{3,4}$. We partition the rows into $\{2, 4, 6\}$ and $\{1, 3, 5, 7\}$ (widths 24, 25, 26, and 28) and into $\{1, 4, 7\}$ and $\{2, 3, 5, 6\}$ (width 27). The resulting blackouts are

$$56, 45, 34, 23, 1, 1, 14, 1, 12, 1, 6, 16, 7, 1, 47, 6, 67, 7, 7, 6, 25, 7, 36, 27,$$
$$16, 25, 7, 27, 1, 7, 7, 7, 12, 2, 14, 1, 1, 23, 34, 45, 56, 7, 7, 47, 6, 6, 4, 36, 67,$$
$$67, 47, 23, 12, 1, 7, 1, 2, 2, 25, 5, 36, 6, 6, 45, 1, 6, 27, 1, 1, 1, 7, 34, 14, 56, 16,$$
$$57, 1, 16, 37, 3, 67, 1, 7, 1, 1, 34, 7, 24, 1, 15, 27, 7, 1, 7, 7, 45, 1, 46, 7, 13, 6, 12,$$
$$34, 2, 2, 2, 47, 2, 2, 27, 25, 7, 7, 7, 7, 7, 67, 36, 23, 2, 12, 1, 1, 16, 1, 1, 14, 1, 45, 56$$

of widths 24 to 28. Three enumerators confirm all five. The last one stays valid after adding $W_7 = 10$ columns black only in row 7 on its left. Theorem 39, with padding row 7 on the left side, then gives $m + 12$ points at every width $m \geq 28$. Three enumerators also confirm this collar test board. □

Theorem 45 (small grids, computer-assisted)**.** *We have $f(6, 6) = 12$, $f(6, 7) = 13$, $f(7, 7) = 14$, and $f(6, 8) = 15$. The blackout $15, 1, 123, 34, 4, 4, 456, 26$ of the 6×8 grid attains the last value.*

Proof. The other constructions are $14, 1, 12, 2, 23, 3456$ on the 6×6 grid, $16, 1, 1, 12, 2, 23, 3456$ on 6×7, and $17, 1, 12, 2, 23, 3456, 67$ on 7×7. Their records are `jis/data/small_cells_sat_6x6.json`, `jis/data/small_cells_sat_6x7.json`, and `jis/data/small_cells_sat_7x7.json`. All are valid. For each grid, a checked SAT certificate excludes one additional point. □

Theorem 46 (the 5×5 grid)**.**

(a) We have $f(5,5) = 10$*.*

(b) (Computer-assisted.) The $a_5(5) = 8$ *maximum blackouts are the images of* S_0 *below under the square's symmetries.*

Proof. (a) Corollary 12 gives the upper bound. To attain it, take S_0 to be $45, 34, 123, 1, 15$. Its row-column graph is an eight-cycle with two pendant edges, so it is axis-valid. Lemma 13 lists 30 tilted rectangles, namely nine unit diamonds, sixteen others from Lemma 16, and five spanning the grid. The fifteen weak rectangles show the pairs below. After an oblique pair, the point following the bar is a white corner of the axis-aligned rectangle through it.

$(1,2),(2,1) \mid (1,1)$	$(1,3),(2,2) \mid (1,2)$	$(3,4),(4,3) \mid (4,4)$	$(2,5),(3,4) \mid (3,5)$
$(2,2),(4,2)$	$(1,2),(4,3) \mid (1,3)$	$(4,4),(5,2) \mid (4,2)$	$(3,5),(4,4) \mid (3,4)$
$(4,3),(5,2) \mid (4,2)$	$(3,4),(5,2) \mid (5,4)$	$(2,5),(5,2) \mid (2,2)$	$(4,5),(5,3) \mid (4,3)$
$(1,3),(4,2) \mid (1,2)$	$(2,5),(4,3) \mid (4,5)$		$(4,5),(5,4) \mid (4,4)$

These pairs are all different. The other tilted rectangles show at least three points, so Lemma 4 distinguishes them. The listed white corners rule out axis-aligned rivals for every oblique pair. The remaining pair $(2,2),(4,2)$ has no rival because $B_2 \cap B_4 = \emptyset$. This proves validity. (b) Three programs check all 44,640 sets of 10 points whose graph has no four- or six-cycle. Only the images of S_0 are valid. □

8 Every height at large widths

We now attain the row bound at every height with one construction. Place the double columns far apart and alternate the intervening black points between the boundary rows. Call a singleton column a *sea* if its black point lies in row 1 or row h.

Theorem 47 (large widths)**.** *Let* $h \geq 3$ *and put* $M_h = q_h + (q_h - 1)(2W_h + 1)$*. Then* $f(h,m) = m + q_h$ *for every* $m \geq M_h$*.*

Theorem 8 gives the upper bound. For the construction, we can use any triangle-free graph on the rows.

Proposition 48 (alternating seas)**.** *Let* $h \geq 3$ *and let* H *be a triangle-free graph on the rows with edges* $e_1, \ldots, e_q$*. Let* Z *be the blackout of the* $h \times m$ *grid with doubles in columns* $p_1 < \cdots < p_q$*, where column* p_j *is black exactly in the two rows of* e_j *and* $p_{j+1} - p_j \geq 2W_h + 2$*, and with a sea in every other column. The seas to the left of* p_1 *are black in row* 1*, and the row of the seas switches between* 1 *and* h *at every double. Then* Z *is valid and* $|Z| = m + q$*.*

Proof of Theorem 47. Take $H = K_{\lfloor h/2 \rfloor, \lceil h/2 \rceil}$, so $q = q_h$, and place the doubles at $p_j = 1 + (j-1)(2W_h + 2)$. The last is in column $1 + (q_h - 1)(2W_h + 2) = M_h$. Every later column is a sea, so Proposition 48 gives $m + q_h$ black points for every $m \geq M_h$. □

We use three facts about a tilted rectangle R of the $h \times m$ grid.

(F1) R has at most one corner in row 1 and at most one in row h, since each extreme row holds only one corner (Lemma 13).

(F2) Corners in rows 1 and h are opposite. If adjacent, their side would be $(s, h-1)$. Translating them by the nonzero perpendicular side n gives the other corners. Staying in the grid requires $n_2 \geq 0$ at row 1 and $n_2 \leq 0$ at row h, so $n_2 = 0$. Then $n_1 s = 0$ forces $s = 0$, contradicting the absence of vertical sides.

(F3) Two corners of R in one column are opposite, since a vertical side would make R axis-aligned.

Proof of Proposition 48. Each double adds one point to a column, so $|Z| = m + q$.

Axis-aligned rectangles. Only doubles have two black rows, and their pairs are distinct, so G_Z has no four-cycle. A six-cycle $r_a\, c_x\, r_b\, c_y\, r_c\, c_z$ would need doubles on $\{a, b\}$, $\{b, c\}$, and $\{a, c\}$, a triangle in H. Thus Z distinguishes axis-aligned rectangles by Lemma 7. None is hidden, since that would give a four-cycle.

Near one double. A tilted rectangle R spans at most W_h (Lemma 14), so its columns meet at most one double. With no double, all black points in those columns lie in one boundary row. By (F1), R then shows at least three corners. Otherwise, shift the double to column 0 and write its black rows as $a < b$. Reflecting rows if needed, take the seas to be black in row 1 to the left of column 0 and row h to the right.

Let K be the black corners of R. By (F1), these can only be $u = (x_1, 1)$ with $x_1 < 0$, $v = (x_2, h)$ with $x_2 > 0$, and points $d = (0, c)$ with $c \in \{a, b\}$. A corner $(0, 1)$ or $(0, h)$ counts as d and excludes u or v, respectively. We need only consider $|K| \geq 2$.

(a) $K = \{u, v\}$. These are opposite by (F2). The other diagonal has midpoint $\mu = \left(\frac{x_1+x_2}{2}, \frac{1+h}{2}\right)$ and endpoints $\mu \pm w$ with $|w|^2 = \frac{1}{4}\big((x_2 - x_1)^2 + (h-1)^2\big)$. By (F1) its endpoints lie in rows 2 to $h-1$, so $|w_2| \leq \frac{h-3}{2}$ and $w_1^2 > \frac{1}{4}(x_2 - x_1)^2$. One endpoint is therefore left of x_1 and the other right of x_2. Neither lies in column 0, so both are white.

(b) $K = \{(0, a), (0, b)\}$. These are opposite by (F3). The other diagonal has midpoint $\big(0, \frac{a+b}{2}\big)$, length $b - a$, and $w_1 \neq 0$, so $|w_2| < \frac{b-a}{2}$. Its endpoints lie in the columns $\pm w_1$ and strictly between rows a and b, so both are white.

(c) $K = \{u, d\}$, opposite. Here $c \geq 2$ by (F1). The other diagonal has midpoint $\left(\frac{x_1}{2}, \frac{1+c}{2}\right)$ and $|w|^2 = \frac{1}{4}\big(x_1^2 + (c-1)^2\big)$. Its endpoints are not in row 1 by (F1), not in row h by (F2), since the opposite corner of u is d, and not in column 0 by (F3). So both are white. Since their rows are at least 2, we get $|w_2| \leq \frac{c-3}{2}$. Thus $c \geq 3$ and $w_1^2 > \frac{1}{4}x_1^2$, so one endpoint lies right of 0 and the other left of x_1.

(d) $K = \{u, d\}$, adjacent. The side $d - u = (|x_1|, c-1)$ has $c \geq 2$, and the other corners are $u + n$ and $d + n$ with $n = t\big(-(c-1), |x_1|\big)$ for some $t \neq 0$. The corner $u + n$ lies in

row $1+t|x_1|$, so $t>0$. Then both corners lie in negative columns and below row 1, so both are white. They differ by $(|x_1|, c-1)$, so they form an oblique pair.

(e) K contains v and a point d but not u. This is (c) or (d) after the half turn $(x,j)\mapsto(-x,h+1-j)$, which exchanges the two seas and fixes column 0.

(f) $|K|\geq 3$. If K contains $(0,a)$ and $(0,b)$, the other two corners lie strictly between rows a and b by (b), so they are neither u nor v. Otherwise $K=\{u,d,v\}$. Then u and v are opposite by (F2), and d is adjacent to both, so $(u-d)\cdot(v-d)=0$, that is, $x_1x_2=(c-1)(h-c)$. The left side is negative and the right side is not, a contradiction.

Thus every weak tilted rectangle shows exactly two corners. In (a) to (c) and their reflections, the pair straddles column 0. In (d) and its reflection, it is oblique and lies on one side of column 0.

Axis-aligned rivals. By Lemma 18, only a horizontal or an oblique pair needs checking. A horizontal pair $\{(x,j),(z,j)\}$ arises only in (a) to (c), with $x<0<z$. There $B_x=\{1\}$ and $B_z=\{h\}$ are disjoint, so it has no rival. For an oblique pair $\{(x,j),(z,k)\}$, a rival needs (x,k) and (z,j) black. In (a) to (c), the rows j,k lie in $2,\dots,h-1$ and $x,z\neq 0$, so these points are white. In (d), $x,z<0$ and $j,k\geq 2$, so again they are white.

Two tilted rectangles. Let weak tilted rectangles $R\neq R'$ show the same pair P. The columns of each contain P and span at most W_h, so their doubles are at most $2W_h$ apart. Hence they share the double in column 0. If P is a diagonal of both, their black diagonals have the same midpoint and the same length. Comparing the cases, these determine the black diagonal. For example, (a) against (c) forces $c=h$ from the midpoint rows, then $x_1'=x_1+x_2$ and $|x_1'|=x_2-x_1$, so $x_1=0$ or $x_2=0$. Case (b) has its midpoint in column 0, and (a) can match it only with $x_1=-x_2$ and $(2x_2)^2+(h-1)^2=(b-a)^2\leq(h-1)^2$, which is impossible. The remaining comparisons use the same midpoint and length constraints. If P is a side of exactly one of them, it lies on one side of column 0 by (d) and on both sides by (a) to (c), a contradiction.

Suppose finally that P is a side of both, so the black sides $Y=P-n$ and $Y'=P-n'$ satisfy $Y'=Y+(n-n')$. If both are of type (d), with $Y=\{u,d\}$ and $Y'=\{u',d'\}$, then matching u with u' gives $n=n'$, since d,d' lie in column 0 and u,u' in row 1. Matching u with d' would put u' in column $-x_1>0$. Two sides of the reflected type are symmetric. If $Y=\{u,d\}$ and $Y'=\{d',v'\}$, matching d with d' would put v' in column $x_1<0$. Matching u with d' gives $n_1-n_1'=|x_1|>0$, but $n_1<0<n_1'$, since the white corners of R lie left of column 0 and those of R' lie right. So $R=R'$.

Presentations of three or more points are unique (Lemma 4), so Z is valid. □

Theorem 47 gives $k_r=q_h$ for every residue in Theorem 50(a), but the threshold M_h is not sharp for small h. It is 61 for $h=5$, whereas $f(5,m)=m+6$ from $m=8$. For $h=4$, 5, 6, and 7, the least widths are 6, 8, 15, and at most 24.

9 Every fixed height

At fixed height h, finite memory suffices to test validity and count optima. Read a blackout as the *column word* $w = B_1 \cdots B_m$, with B_i recording column i's black rows. Write $|w| = m$ and $S(w)$ for its blackout, and let $\mathcal{L}_h$ be the valid words, including the empty word. We first recognize all valid words, then add a bounded counter to select optima.

Theorem 49 (regularity and rational series)**.** *For every $h \geq 1$ the language $\mathcal{L}_h$ is regular, and $G_h(z,u) = \sum_{w \in \mathcal{L}_h} z^{|w|} u^{|S(w)|}$ is rational in z and u.*

Proof. The *trace* of a rectangle records, in each column, the rows of its corners. It determines the corner set. By Lemma 2, two distinct traces Y, Y' give the same presentation exactly when $Y_i \triangle Y_i' \subseteq B_i$ in every column.

The possible traces form a regular language. For an axis-aligned rectangle, choose $r < s$, read a column $\{r, s\}$, then another such column after any number of empty columns. There can also be empty columns before and after. For a tilted rectangle, Lemmas 13 and 14 leave only finitely many templates (1), each of span at most W_h, again with arbitrary empty columns on either side.

An automaton can therefore read w while guessing two rectangle traces. It checks the containment in each column and records whether the traces ever differ. Accepting exactly when they differ recognizes the invalid words. Closure under complementation then makes $\mathcal{L}_h$ regular.

Choose a deterministic automaton for $\mathcal{L}_h$ and give a transition reading B the weight $u^{|B|}$. If $M(u)$ is its weighted transition matrix, α its start vector and β its accepting vector, then

$$G_h(z,u) = \alpha(I - zM(u))^{-1}\beta.$$

Each word is counted once because the automaton is deterministic. Cramer's rule makes the expression rational [11, Proposition V.6]. □

Theorem 50 (fixed height)**.** *Let $h \geq 2$. (a) There are integers $m_0 \geq 3$, $p \geq 1$, and $k_0, \ldots, k_{p-1}$ such that $f(h,m) = m + k_{m \bmod p}$ for every $m \geq m_0$. (b) The series $\sum_{m \geq 0} a_h(m) z^m$ is rational, with the conventions of Section 2, and the words of optimum blackouts form a regular language.*

Proof. Put $E(w) = |S(w)| - |w|$, the number of black points beyond one per column. Every consecutive block v of a valid word is valid on its own grid. Theorem 8 bounds $E(v)$ by q_h when $|v| \geq 3$. For a shorter block, $|S(v)| \leq h|v|$ gives the common bound

$$E(v) \leq C_h := \max(q_h, 2h - 2).$$

If a valid word $w = uv$ has excess k, applying this to the prefix and suffix gives

$$k - C_h \leq E(u) = k - E(v) \leq C_h.$$

For fixed k, the counter therefore needs only finitely many values. Add it to an automaton for $\mathcal{L}_h$ and accept when the final excess is k. This recognizes the regular language $\mathcal{L}_{h,k}$.

The set of lengths of a regular language is eventually periodic. For $m \geq 3$, Lemma 17 and Theorem 8 put the optimal excess between $h-1$ and q_h. These finitely many k admit a common period p and threshold m_0 for their length sets. The largest available excess depends only on $m \bmod p$. Calling it k_r proves (a).

For each residue r, keep the words of $\mathcal{L}_{h,k_r}$ whose lengths are at least m_0 and congruent to r modulo p. Each resulting language is regular. Their union and the finitely many shorter optimum words give all optimum blackouts. The transfer-matrix argument makes their counting series rational, proving (b). □

10 Square grids

On square grids, $f(n,n)$ eventually exceeds every fixed multiple of n. We prove this by choosing black points at random, then deleting points until all rectangle presentations differ.

Fix $n \geq 2$ and put $\Lambda = 1 + \ln n$. Write $\mathcal{R}$ for all rectangles on the $n \times n$ grid, including tilted ones, and d_v for the number through corner v.

Lemma 51. *If rectangles $R_1 \neq R_2$ have exactly two common corners p and q, and $\{p,q\}$ is a side of both or a diagonal of both, then $C(R_1) \triangle C(R_2)$ is the corner set of a rectangle.*

Proof. For a common side, the remaining corners $p+u, q+u$ and $p+v, q+v$ have $u \neq v$ nonzero and perpendicular to $q-p$. They form a rectangle with sides $q-p$ and $v-u$. For a common diagonal, write them as $o \pm u, o \pm v$, with $o = (p+q)/2$, $|u| = |v|$, and $v \neq \pm u$. The equal segments $[o-u, o+u]$ and $[o-v, o+v]$ share a midpoint, so they are diagonals of a rectangle. □

Let $r_2(t)$ be the number of integer pairs (a,b) satisfying $a^2+b^2 = t$, and $\tau(t)$ the divisor count of t. We use the estimates $r_2(t) \leq 4\tau(t)$ and, for each $\varepsilon > 0$, $\tau(t) \leq K_\varepsilon t^\varepsilon$ for $t \geq 1$, with a constant K_ε [16, Sections 16.9, 16.10, and 18.1].

Collect two kinds of four-point obstructions in $\mathcal{A}$, namely the full corner sets $C(R)$ for $R \in \mathcal{R}$, and the differences $C(R_1) \triangle C(R_2)$ when exactly two shared corners are a side of one rectangle and a diagonal of the other. Let $\mathcal{B}$ contain the differences $C(R_1) \triangle C(R_2)$ for pairs with exactly one shared corner.

Lemma 52.

(a) Each point v has $d_v \leq 8n^2\Lambda$, and $|\mathcal{R}| \leq 2n^4\Lambda$.

(b) For every $\varepsilon > 0$, $|\mathcal{A}| \leq A_\varepsilon n^{4+2\varepsilon}\Lambda$ with $A_\varepsilon = 2(1 + 16 \cdot 2^\varepsilon K_\varepsilon)$, and $|\mathcal{B}| \leq 32n^6\Lambda^2$.

(c) A blackout that contains no member of $\mathcal{A} \cup \mathcal{B}$ is valid.

Proof. (a) Write the sides at v as $k(a,b)$ and $l(-b,a)$, with $\gcd(a,b) = 1$, $k \geq 1$, and $l \neq 0$. For $d = \max(|a|,|b|)$, there are at most $8d$ primitive directions, at most $(n-1)/d$ choices for k, and $2(n-1)/d$ for l. Each rectangle is counted twice, so $2d_v \leq \sum_{d=1}^{n-1} 16(n-1)^2/d \leq 16n^2\Lambda$. Sum over corners to get $4|\mathcal{R}| = \sum_v d_v \leq 8n^4\Lambda$. (b) With diagonal $\{p,q\}$ fixed, the other

diagonal determines the rectangle. It is an integer vector with squared length $t = |p - q|^2 < 2n^2$. Hence a side of any $R \in \mathcal{R}$ is the diagonal of at most $r_2(t) \le 4K_\varepsilon(2n^2)^\varepsilon$ rectangles. Counting the four sides gives $|\mathcal{A}| \le |\mathcal{R}|(1 + 16K_\varepsilon 2^\varepsilon n^{2\varepsilon}) \le A_\varepsilon n^{4+2\varepsilon}\Lambda$. For $\mathcal{B}$, count the pairs of rectangles through a common corner, giving $|\mathcal{B}| \le \sum_v \binom{d_v}{2} \le \frac{n^2}{2}(8n^2\Lambda)^2$. (c) An invalid S contains $C(R_1) \triangle C(R_2)$ for some $R_1 \ne R_2$ by Lemma 2. They share at most two corners (Lemma 4). With two shared corners, the difference belongs to $\mathcal{A}$, either by Lemma 51 or by definition. With one, it belongs to $\mathcal{B}$. With none, $C(R_1) \subseteq S$ itself belongs to $\mathcal{A}$. □

Theorem 53 (deletion method)**.** *We have*

$$\liminf_{n\to\infty} \frac{f(n,n)\,(1+\ln n)^{2/5}}{n^{6/5}} \ge \frac{5}{6 \cdot 192^{1/5}} = 0.2911\ldots.$$

In particular $f(n,n) - 2n \to \infty$.

Proof. Use the deletion method of Alon and Spencer [2, Section 3.2]. Set $a = 192^{-1/5}$, $T = n^{6/5}\Lambda^{-2/5}$, and $\theta = a\,n^{-4/5}\Lambda^{-2/5} \in (0,1)$. Choose each point independently with probability θ. Let X count chosen points, and let Y and Z count fully chosen members of $\mathcal{A}$ and $\mathcal{B}$, respectively. Then

$$\mathbb{E}(X - Y - Z) = n^2\theta - |\mathcal{A}|\theta^4 - |\mathcal{B}|\theta^6.$$

Here $n^2\theta = aT$. From Lemma 52(b), with $\varepsilon = 1/10$, the two losses are bounded by $|\mathcal{B}|\theta^6 \le 32a^6T$ and $|\mathcal{A}|\theta^4 \le A_{1/10}a^4n^{-1/5}T$. Since $32a^5 = \frac{1}{6}$, some chosen set S' satisfies $X - Y - Z \ge (\frac{5}{6}a - A_{1/10}a^4n^{-1/5})T$. Delete one point from each member of $\mathcal{A} \cup \mathcal{B}$ contained in S'. We lose at most $Y + Z$ points, leaving at least $X - Y - Z$ in a valid blackout by Lemma 52(c). Consequently $f(n,n)/T \ge \frac{5}{6}a - A_{1/10}a^4n^{-1/5} \to 5/(6 \cdot 192^{1/5})$. Finally $T/n \to \infty$, so $f(n,n) - 2n \to \infty$. □

The exponent $6/5$ comes from deleting about n^6 six-point obstructions. It falls short of the upper bound $n^{4/3}$ in Section 12. For axis-aligned rectangles, Lemma 7 requires only a row-column graph without four- or six-cycles. Incidence graphs of generalized quadrangles give such blackouts of order $n^{4/3}$ [5].

The difficulty is to retain that size while distinguishing tilted rectangles. A random set of density $n^{-2/3}$ has about $n^6 \cdot n^{-4} = n^2$ members of $\mathcal{B}$ in expectation, up to logarithmic factors, but only $n^{4/3}$ points. Girth eight excludes only those members arising from two axis-aligned rectangles. To reach order $n^{4/3}$, this suggests seeking constructions whose algebraic structure also controls the tilted rectangles.

11 Computations and verification

Every computed maximum and exact count has two separately written checks. The bound $a_7(8) \ge 18$ comes from one enumeration. The repository [31] stores the code and evidence, with commands, SHA-256 hashes, and running times in `jis/scripts/README.md`. All paths

start from `kagey-problems/problem001`. The supplement tabulates the values and lists each computer-assisted claim's checks and evidence files.

We solve the hitting-set problem of Lemma 2 with CBC [12] through PuLP [28], and check the ILP results by brute force. SAT enumeration uses PySAT [18]. The column searches grow the board from left to right. They use $f(6,6) \le 12$ to prune 6×7 and 6×8, and $f(6,7) \le 13$ to prune 6×8 and 7×7. Every saved construction and counted model is tested for validity.

Encoding A forbids an entirely black inclusion-minimal difference $C(R_1) \triangle C(R_2)$, $R_1 \neq R_2$. Encoding B forbids $C(R_1) \triangle C(R_2)$ only for distinct rectangles with a common corner. Each rectangle R also has an indicator y_R, forced true if it is hidden, and at most one y_R may be true.

For a structured six-row instance, fix a row graph from Theorem 11 or Lemma 42. A column's type is its black-row set, either a singleton or an edge, with the empty set also allowed in case (c). Use each edge once, and the empty type once in case (c). Of encoding B's clauses, retain those involving a tilted rectangle whose forbidden set can be all black. Cardinality is enforced by Sinz's sequential counter [33], expanded by our generator for certified instances, or by a totalizer [3]. Both enforce exactly the specified size bound.

Lemma 54 (faithful encodings)**.** *Let S be a blackout of the $h \times m$ grid, and let $\mathcal{D} = \{C(R_1) \triangle C(R_2) \mid R_1 \neq R_2\}$, where R_1 and R_2 range over rectangles of this grid.*

(a) S is valid if and only if it contains no inclusion-minimal member of $\mathcal{D}$.

(b) S is valid if and only if it hides at most one rectangle and contains no $C(R_1) \triangle C(R_2)$ for distinct rectangles R_1, R_2 sharing a corner.

(c) Suppose $h = 6$, $m \ge 3$, and S is valid with $m+9$ or $m+8$ points. For one of the graphs of Theorem 11 or Lemma 42, selecting the black-row set B_i as the type of column i extends to a satisfying assignment of the structured instance.

Proof. (a) Since $\mathcal{D}$ is finite, each member contains an inclusion-minimal one. Excluding these therefore excludes every forbidden difference. Apply Corollary 3.

(b) It remains to check pairs with no shared corner. Their difference is $C(R_1) \cup C(R_2)$, contained in S exactly when both rectangles are hidden. Allowing at most one hidden rectangle excludes precisely these pairs.

(c) Theorem 11 or Lemma 42 gives the graph. Every B_i is a singleton, an edge of it, or, in case (c) of the lemma, empty, and every edge occurs once. Use these as the column types. Set y_R to 1 exactly for hidden rectangles. By (b), these choices satisfy encoding B. Dropping clauses preserves satisfaction in the structured instance. □

The point variables of A or B therefore describe exactly the valid blackouts of the chosen size. For six rows, part (c) and the row bound imply $f(6,m) \le m+8$ once the ten structured instances of width m are all unsatisfiable, and $f(6,m) \le m+7$ once the further 115 are.

CaDiCaL [6] produces LRAT certificates for every SAT upper bound. These cover $f(5,6) < 12$, $f(5,7) < 13$, $f(6,6) < 13$, $f(6,7) < 14$, $f(7,7) < 15$, $f(6,8) < 16$, and

$f(7,8) < 17$ using encoding A on all blackouts, the 70 structured six-row instances with $m+9$ points for widths 8 to 14, and the 345 with $m+8$ points for widths 9 to 11. The cell instances on 6×8, 7×7, and 7×8 are split by their first column. Every one of the 739 proofs passes both drat-trim's `lrat-check` [7, 36] and the Lean LRAT checker.

The formal proofs are in `../lean/KageyProblems/Problem001`. They cover Lemmas 2, 4, and 14, Corollary 3, the strip theorem, both orientations of the row bound, the collar theorem, and Theorem 47 with Proposition 48 in full, for every $h \geq 3$. Kernel-checked validity tests of the constructions and collar boards then prove $f(5,m) = m+6$ for $m \geq 8$, $f(6,m) = m+9$ for $m \geq 15$, $f(5,5) \geq 10$, and $f(6,8) \geq 15$ from only Lean's three standard axioms. The supplement lists these tests and the remaining unformalized parts.

12 Open questions

(1) What are the rational series $\sum_{m \geq 0} a_5(m) z^m$ from Theorem 50 and the growth rate of $a_5(m)$? Remark 41 gives the known counts.

(2) Let m_h be the least width from which $f(h,m) = m + q_h$. Theorem 47 gives $m_h \leq q_h + (q_h - 1)(2W_h + 1)$, of order h^4, while $m_4 = 6$, $m_5 = 8$, $m_6 = 15$, and $m_7 \leq 24$. What is the order of growth of m_h, and what is m_7?

(3) Lemma 7 and Neuwirth's bound for bipartite graphs of girth eight [30, Corollary 16] give $f(n,n) < n^{4/3} + \frac{2}{3}n - \frac{2}{9}n^{2/3} - \frac{20}{81}n^{1/3}$ for $n \geq 3$, and Theorem 53 gives $f(n,n) \geq 0.29\, n^{6/5}(1 + \ln n)^{-2/5}$ for large n. Lavrov [20] used the same method for the triangular version. What is the true order of growth? What is the first n with $f(n,n) > 2n$?

(4) Corollary 15 settles the largest entry in each row of A338885. Is its smallest entry in row n equal to A228286(n), as conjectured there?

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A note on the use of AI

I began this work in January of 2026 and wrote the first draft and its mathematics between August and October, together with most of the Python code. I used Claude Opus 5.5 (Anthropic) for the October 17–19 revision, including the cut of a draft from 45 to 30 pages and the language passes. It did not add any language, just proposed a cut version. Using this as reference, I rewrote the text myself and implemented the revisions as I saw fit. No

sentence in this paper was written by an LLM. It also drew the figures in the style of Kagey's posting of the problem given my own hand-drawn sketches.

I used it to check my Python code, and it wrote further checks of the computer-assisted results. One of these, `six_row_collar_6x9_check.py`, gives the lower bound in Theorem 43(a). It also wrote the Lean files, the LRAT certificate pipeline and the formal checks described in Theorem A and Section 11. The formalization did not change a statement of the paper or find an error or missing hypothesis. It changed how several results are established, as Section 11 describes.

I also used Claude Opus 5.5 to search the literature beyond the sources I had found. Several references came from that search. I read every cited source, including the parts supporting each citation. Later reviews with Claude Opus 5.5 and OpenAI Codex on October 23rd pointed out problems that I checked and corrected, such as typos and inconsistencies or redundancies. I checked these revisions and implemented them all myself, and take full responsibility for the contents of the paper.

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